

Question number	Answer	Additional guidance	Mark
5(a)(i)	Choose an item. • B is the correct answer A is not the correct answer as glucose and prothrombin are not filtered from the blood in the renal capsule C is not the correct answer as glycogen and urea are not filtered from the blood in the renal capsule D is not the correct answer as urea and prothrombin are not filtered from the blood in the renal capsule		Computer (1)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	Choose an item. • A is the correct answer B is not the correct answer as the endothelial cells do not prevent plasma proteins being filtered from the blood stream C is not the correct answer as the epithelial cells do not prevent plasma proteins being filtered from the blood stream D is not the correct answer as epithelial cells of the loop of Henlé do not prevent plasma proteins being filtered from the blood stream		Computer (1)

Question number	Answer	Additional guidance	Mark
5(a)(iii)	Choose an item: • D is the correct answer A is not the correct answer as endocytosis is not the glucose transport mechanism B is not the correct answer as exocytosis is not the glucose transport mechanism C is not the correct answer as osmosis is not the glucose transport mechanism		Computer (1)

5(b)	A description that includes four of the following points: • {fluid / filtrate} (in the two limbs of the loop of Henle) moves in opposite directions (1) • (sodium / potassium / chloride) ions move out of ascending limb (into surrounding fluid) (1) • (ions move) by active transport (1) • (resulting in) increase in {(ion) concentration / decrease in water potential} (of surrounding fluid) (1) • (therefore) water leaves {the descending limb / by osmosis / through aquaporins} (1)	accept symbols for ions eg Na⁺ do not accept sodium accept idea of active transport – / actively pumped / pumped / moved actively can get this even if the ions were pumped in wrong direction accept (surrounding) fluid becomes hypertonic accept a (water / ion / solute) concentration gradient formed accept there is a greater (solute) concentration outside the base of the descending limb accept diffusion of water accept (because) the descending limb is permeable to water accept the ascending limb is impermeable to water	Expert (4)

Question number	Answer	Additional guidance	Mark
5(c)(i)	A calculation showing the following steps: - correct readings from the graph for the three sets of females put into a ratio (1) - calculation of ratio to 1 significant figure (1)	100:840:440 in working = 1 mark 1:8:4 = 2 marks 1:8.4:4.4 = 1 mark	Graduate (2)

Question number	Answer	Additional guidance	Mark
5(c)(ii)	A description that includes the following points: {drinking leads to higher urine production / not drinking leads to least urine production} (1) {urine production is higher in females when they have had a drink / the increase in females is greater than in males}(1)	either water or salt solution or just drinking applies to both males and females or neither NOT just one of them ignore females produce more urine than males (unqualified) accept converse for males accept group letters B and C for drinking	Expert (2)

Question number	Answer	Additional guidance	Mark
5(c)(iii)	An explanation that includes two of the following points: - increased water {potential / concentration} (of blood) (1) - ADH not released / less ADH released (1) - decreased permeability of walls of {distal convoluted tubule / collecting duct} to water (1) - less reabsorption of water (into blood) (1)	ALLOW reference to detected by osmoreceptors allow increased water in blood ignore loop of Henle	Expert (2)

